Supplementary Material for "Instance Optimal Sparse Recovery from Nonlinear Observations: A Unified Framework"

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September 2, 2026

Abstract

This document is the supplementary material for [2], which developed a unified framework for instance optimal sparse recovery from nonlinear observations. In Section 1, we show that the framework readily recovers the instance optimality (of iterative hard thresholding) in classical compressed sensing. In Section 2, we show that the framework applies to phase-only compressed sensing and yields instance optimal guarantee (of a novel algorithm) comparable to [4]. To be concrete, we consider Gaussian designs.

1 Compressed Sensing

We consider sparse recovery from

$$\mathbf{y} = \mathbf{A}\mathbf{x}$$

with Gaussian sensing matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$, i.e., compressed sensing with Gaussian design. Note that $\bar{\mathcal{X}} = \mathbb{R}^n$, and it turns out that we can take $\mathcal{X} = \bar{\mathcal{X}}$. For each $\mathbf{x} \in \mathbb{R}^n$, we adopt ℓ_2 loss function

$$\mathcal{L}_{\mathbf{x}}(\mathbf{u}) = \frac{1}{2m}\|\mathbf{A}\mathbf{u} - \mathbf{y}\|_2^2 = \frac{1}{2m}\|\mathbf{A}(\mathbf{u}-\mathbf{x})\|_2^2$$

whose gradient is

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) = \frac{1}{m}\mathbf{A}^\top\mathbf{A}(\mathbf{u}-\mathbf{x}).$$

1.1 Instance Optimality

Lemma 1.1. *Under $m \gtrsim s\log(\frac{en}{s})$,*

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \mathrm{RAIC}\Big(\Sigma_{2s}^{n,*}; \Sigma_s^n, \mathbf{x}, \frac{1}{5}\|\mathbf{u}-\mathbf{x}\|_2 + \frac{2}{5}\sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2\Big),\ \forall \mathbf{x}\in\mathbb{R}^n \tag{1.1}$$

holds with probability at least $1 - 2\exp(-cm)$.

Proof. The desired (1.1) is equivalent to

$$\Big\|\mathbf{u} - \mathbf{x} - \frac{1}{m}\mathbf{A}^\top\mathbf{A}(\mathbf{u}-\mathbf{x})\Big\|_{2s,2} \leq \frac{1}{5}\Big(\|\mathbf{u}-\mathbf{x}\|_2 + 2\sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2\Big) \tag{1.2}$$

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for all $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{u} \in \Sigma_s^n$. By $\mathbf{u} - \mathbf{x} = \mathbf{u} - \mathbf{x}_{[s]} - \sum_{i\geq 1}(\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]})$ and triangle inequality, we start with

$$\begin{aligned}
&\left\| \mathbf{u} - \mathbf{x} - \frac{1}{m}\mathbf{A}^\top\mathbf{A}(\mathbf{u}-\mathbf{x}) \right\|_{2s,2} \\
&\leq \left\| \left(\mathbf{I}_n - \frac{\mathbf{A}^\top\mathbf{A}}{m}\right)(\mathbf{u} - \mathbf{x}_{[s]}) \right\|_{2s,2} + \sum_{i\geq 1} \left\| \left(\mathbf{I}_n - \frac{\mathbf{A}^\top\mathbf{A}}{m}\right)(\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}) \right\|_{2s,2}.
\end{aligned} \tag{1.3}$$

A straightforward application of [2, Lemma C.13] (concentration of product process) yields that with probability at least $1 - \exp(-cm)$,

$$\left\| \left(\mathbf{I}_n - \frac{\mathbf{A}^\top\mathbf{A}}{m}\right)\mathbf{v} \right\|_{2s,2} = \sup_{\mathbf{w}\in\Sigma_{2s}^{n,*}} \left| \frac{\langle \mathbf{A}\mathbf{w}, \mathbf{A}\mathbf{v}\rangle}{m} - \langle \mathbf{w}, \mathbf{v}\rangle \right| \leq \frac{\|\mathbf{v}\|_2}{5}, \quad \forall \mathbf{v} \in \Sigma_{2s}^n. \tag{1.4}$$

Applying (1.4) to (1.3) yields that, for all $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{u} \in \Sigma_s^n$,

$$\left\| \mathbf{u} - \mathbf{x} - \frac{1}{m}\mathbf{A}^\top\mathbf{A}(\mathbf{u}-\mathbf{x}) \right\|_{2s,2} \leq \frac{1}{5}\left(\|\mathbf{u} - \mathbf{x}_{[s]}\|_2 + \sum_{i\geq 1} \left\|\mathbf{x}_{(i+1)s} - \mathbf{x}_{[is]}\right\|_2 \right).$$

Further using

$$\|\mathbf{u} - \mathbf{x}_{[s]}\|_2 \leq \|\mathbf{u} - \mathbf{x}\|_2 + \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 \leq \|\mathbf{u} - \mathbf{x}\|_2 + \sum_{i\geq 1} \left\|\mathbf{x}_{(i+1)s} - \mathbf{x}_{[is]}\right\|_2$$

completes the proof. □

Therefore, we can invoke the framework in [2, Section 2] with $R_{\mathbf{x}}^{\mathrm{loc}} = \infty$ and $\mathbf{x}_0 = 0$. Then it yields the following result, which is comparable to [1].

Theorem 1.1 (Instance optimal compressed sensing). *For $\mathbf{x} \in \mathbb{R}^n$, suppose that we obtain $\{\mathbf{x}_t\}_{t\geq 0}$ by running*

$$\mathbf{x}_{t+1} = H_s\left(\mathbf{x}_t - \frac{1}{m}\mathbf{A}^\top(\mathbf{A}\mathbf{x}_t - \mathbf{y}) \right), \quad t = 0, 1, \cdots \tag{1.5}$$

with $\mathbf{x}_0 = 0$. Under Gaussian matrix $\mathbf{A}$ with $m \gtrsim s\log(\frac{en}{s})$, with probability at least $1 - 2\exp(-cm)$,

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \left(\frac{2}{5}\right)^t \|\mathbf{x}\|_2 + \frac{19}{3}\sum_{i\geq 1} \left\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\right\|_2, \quad \forall t \geq 0,\ \forall \mathbf{x} \in \mathbb{R}^n.$$

In particular, if s is even and $s = 2s_0$, then with the same probability,

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \left(\frac{2}{5}\right)^t \|\mathbf{x}\|_2 + \frac{19}{3}\frac{\|\mathbf{x} - \mathbf{x}_{[s_0]}\|_1}{\sqrt{s_0}}, \quad \forall t \geq 0,\ \forall \mathbf{x} \in \mathbb{R}^n. \tag{1.6}$$

1.2 Nonuniform Instance Optimality

Lemma 1.2. *For any fixed $\mathbf{x} \in \mathbb{R}^n$, under $m \gtrsim s\log(\frac{en}{s})$*

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \mathrm{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^n, \mathbf{x}, \frac{1}{5}\|\mathbf{u} - \mathbf{x}\|_2 \right) \tag{1.7}$$

holds with probability at least $1 - 2\exp(-cm)$.

Proof. For fixed $\mathbf{x} \in \mathbb{R}^n$, (1.7) is equivalent to

$$\begin{aligned}
&\left\| \mathbf{u} - \mathbf{x} - \frac{1}{m}\mathbf{A}^\top \mathbf{A}(\mathbf{u}-\mathbf{x}) \right\|_{2s,2} \le \frac{1}{5}\|\mathbf{u}-\mathbf{x}\|_2, \quad \forall \mathbf{u} \in \Sigma_s^n \\
\iff &\left\| \Big(\mathbf{I}_n - \frac{1}{m}\mathbf{A}^\top\mathbf{A}\Big)\frac{\mathbf{u}-\mathbf{x}}{\|\mathbf{u}-\mathbf{x}\|_2} \right\|_{2s,2} \le \frac{1}{5}, \quad \forall \mathbf{u} \in \Sigma_s^n \\
\iff &\sup_{\mathbf{w}\in\Sigma_{2s}^{n,*}} \sup_{\mathbf{v}\in(\Sigma_s^n - \mathbf{x})\setminus\{0\}} \mathbf{w}^\top \Big(\mathbf{I}_n - \frac{1}{m}\mathbf{A}^\top\mathbf{A}\Big)\frac{\mathbf{v}}{\|\mathbf{v}\|_2} \le \frac{1}{5},
\end{aligned}$$

and therefore can be implied by

$$\sup_{\mathbf{w}\in\Sigma_{2s}^{n,*}} \sup_{\mathbf{v}\in(\Sigma_s^n - \mathbb{R}\mathbf{x})\cap\mathbb{S}^{n-1}} \mathbf{w}^\top \Big(\mathbf{I}_n - \frac{1}{m}\mathbf{A}^\top\mathbf{A}\Big)\mathbf{v} \le \frac{1}{5}. \tag{1.8}$$

A straightforward application of [2, Lemma C.13] (concentration of product process) yields (1.8) and completes the proof. □

By the unified framework with $R_{\mathbf{x}}^{\text{loc}} = \infty$ and $\mathbf{x}_0 = 0$, it yields the following nonuniform instance optimality.

Theorem 1.2 (Non-uniform instance optimal compressed sensing)**.** *Assume that we obtain $\{\mathbf{x}_t\}_{t\ge 0}$ by running (1.5) and $\mathbf{x}_0 = 0$. Under Gaussian matrix $\mathbf{A}$, if $m \gtrsim s\log(\frac{en}{s})$, then with probability at least $1-\exp(-cm)$,*

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \le \Big(\frac{2}{5}\Big)^t \|\mathbf{x}\|_2 + 5\|\mathbf{x}-\mathbf{x}_{[s]}\|_2, \quad \forall t \ge 0.$$

2 Phase-Only Compressed Sensing

Recently, the problem of sparse recovery from phase-only measurements receives some attention. The problem of phase-only compressed sensing (PO-CS) concerns the recovery of $\mathbf{x} \in \mathbb{S}^{n-1}$ with a sparse prior from $\mathbf{z} = \text{sign}(\mathbf{\Phi x})$ with complex Gaussian matrix $\mathbf{\Phi} = [\mathbf{\Phi}_1, \cdots, \mathbf{\Phi}_m]^* \in N^{m\times n}(0,1) + N^{m\times n}(0,1)\mathbf{i}$, where the phase of a complex number is defined by $\text{sign}(a) = \frac{a}{|a|}$ for $a \ne 0$ and $\text{sign}(0) = 1$ (by convention). The measurements can be equivalently written as

$$z_i = \text{sign}(\mathbf{\Phi}_i^* \mathbf{x}), \quad i = 1, \cdots, m. \tag{2.1}$$

It was first proved by [6] that exact recovery of $\mathbf{x} \in \mathbb{S}^{n-1}$ can be achieved by a linearization approach. The instance optimality and phase transition were obtained in follow-up works [4, 3]. Particularly, [4] showed that the estimator proposed by [6], denoted by $\hat{\mathbf{x}}_{\text{lin}}$, achieves

$$\|\hat{\mathbf{x}}_{\text{lin}} - \mathbf{x}\|_2 \le \frac{C\|\mathbf{x}-\mathbf{x}_{[s]}\|_1}{\sqrt{s}}, \qquad \forall \mathbf{x} \in \mathbb{S}^{n-1}$$

w.h.p. under $m \gtrsim s\log(en/s)$.

In this section, we shall develop a new algorithm as an instance of normalized iterative hard thresholding (NIHT) and establish an identical result, along with a novel non-uniform (ℓ_2, ℓ_2) instance optimality.

2.1 Algorithm

Notice that (2.1) implies the $\Im(z_i^*\mathbf{\Phi}_i^*\mathbf{x})=0$, we shall consider the ℓ_2 loss

$$\mathcal{L}_{\mathbf{x}}(\mathbf{u})=\frac{1}{2m}\sum_{i=1}^m\left[\Im(z_i^*\mathbf{\Phi}_i^*\mathbf{u})\right]^2=\frac{1}{2m}\sum_{i=1}^m\left[\Im\big(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*\mathbf{u}\big)\right]^2 \tag{2.2}$$

and thus take the gradient

$$\begin{aligned}\mathbf{h}_{\mathbf{x}}(\mathbf{u})=\nabla\mathcal{L}_{\mathbf{x}}(\mathbf{u})&=\frac{1}{m}\sum_{i=1}^m\Im(z_i^*\mathbf{\Phi}_i^*)^\top\Im(z_i^*\mathbf{\Phi}_i^*)\mathbf{u}\\&=\frac{1}{m}\sum_{i=1}^m\Im\big(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*\big)^\top\Im\big(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*\big)\mathbf{u}\\&:=\frac{1}{m}\sum_{i=1}^m\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{u}.\end{aligned} \tag{2.3}$$

by letting $\mathbf{a}_{i,\mathbf{x}}:=\Im\big(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*\big)^\top$.

NIHT. Since $\mathbf{x}\in\mathbb{S}^{n-1}$, we shall consider NIHT:

$$\mathbf{x}_{t+1}=\frac{H_s\big(\mathbf{x}_t-\frac{1}{m}\sum_{i=1}^m\Im(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*)^\top\Im(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*)\mathbf{x}_t\big)}{\big\|H_s\big(\mathbf{x}_t-\frac{1}{m}\sum_{i=1}^m\Im(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*)^\top\Im(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*)\mathbf{x}_t\big)\big\|_2},\quad t\geq 0. \tag{2.4}$$

Initialization. As we shall see, we are only able to prove a "local" RAIC with $R_{\mathbf{x}}^{\mathrm{loc}}$ being a small constant. Thus, we shall construct an accurate initialization. Motivated by the projected back-projection (PBP) method in [5], we adopt

$$\mathbf{x}_0=\frac{H_s(\Re(\frac{1}{m}\sum_{i=1}^m z_i\mathbf{\Phi}_i))}{\|H_s(\Re(\frac{1}{m}\sum_{i=1}^m z_i\mathbf{\Phi}_i))\|_2}. \tag{2.5}$$

Combining (2.4) and (2.5), our complete algorithm is outlined in Algorithm 1. We shall refer to it as normalized phase-only iterative hard thresholding (`NPOIHT`) to parallel `NBIHT` for one-bit compressed sensing.

Algorithm 1 Normalized Phase-Only Iterative Hard Thresholding (`NPOIHT`)

Input: Matrix $\mathbf{\Phi}$, measurements $\mathbf{z}=\mathrm{sign}(\mathbf{\Phi}\mathbf{x})$, sparsity level s

1: **Step 1: Initialization**
2: Compute $\mathbf{x}_0$ from (2.5)
3: **Step 2: Refinement**
4: **for** $t=0,1,2,\dots$ **do**
5: $\quad\mathbf{x}_{t+1}=\dfrac{H_s\big(\mathbf{x}_t-\frac{1}{m}\sum_{i=1}^m\Im(\overline{z_i}\mathbf{\Phi}_i^*)^\top\Im(\overline{z_i}\mathbf{\Phi}_i^*)\mathbf{x}_t\big)}{\big\|H_s\big(\mathbf{x}_t-\frac{1}{m}\sum_{i=1}^m\Im(\overline{z_i}\mathbf{\Phi}_i^*)^\top\Im(\overline{z_i}\mathbf{\Phi}_i^*)\mathbf{x}_t\big)\big\|_2}$,
6: **end for**

Output: Sequence $\{\mathbf{x}_t\}_{t\geq 0}$

Remark 2.1. *We note that our algorithm is novel. In fact, the loss function (2.2) lacks restriction on the signal norm, in light of $\mathcal{L}_{\mathbf{x}}(\lambda\mathbf{x}) = 0$ for all $\lambda \in \mathbb{R}$. To address this issue, the idea of existing works (initiated by [6]) is to add a virtual measurement $\Re(\mathbf{z}^*\mathbf{\Phi}\mathbf{u}) = \sqrt{\frac{\pi}{2}}m$ in order to specify the signal norm. In contrast, we leverage a nonconvex optimization approach and achieve the same goal by normalization.*

2.2 Instance Optimality

We shall first announce our main result for phase-only compressed sensing. Note that it is comparable to [4, Theorem 3.2].

Theorem 2.1 (Instance optimal PO-CS)**.** *For any $\mathbf{x} \in \mathbb{S}^{n-1}$, suppose that $\{\mathbf{x}_t\}_{t\geq 0}$ is generated by running Algorithm 1. Under $\mathbf{\Phi} \sim N^{m\times n}(0,1) + N^{m\times n}(0,1)\mathbf{i}$, if $m \gtrsim s\log(\frac{en}{s})$, then with probability at least $1-\exp(-cm)$,*

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \left(\frac{4}{5}\right)^t + 4000\sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2, \quad \forall t\geq 0,\ \mathbf{x}\in\mathbb{S}^{n-1}.$$

In particular, if s is even and $s = 2s_0$, then with the same probability,

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \left(\frac{4}{5}\right)^t + \frac{4000\|\mathbf{x} - \mathbf{x}_{[s_0]}\|_1}{\sqrt{s_0}}, \quad \forall t\geq 0,\ \mathbf{x}\in\mathbb{S}^{n-1}.$$

Proof. The proof appears in Section 2.4. □

Note that no attempt is made here to optimize the constants.

2.3 Non-Uniform Instance Optimality

We also obtain the following non-uniform instance optimality.

Theorem 2.2 (Non-uniform instance optimal PO-CS)**.** *Given $1 \leq s \leq n$ and fixed $\mathbf{x} \in \mathbb{S}^{n-1}$. Under complex Gaussian $\mathbf{\Phi}$, if $m \gtrsim s\log(\frac{en}{s})$, then with probability at least $1-\exp(-cm)$,*

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq \left(\frac{4}{5}\right)^t\|\mathbf{x}_0 - \mathbf{x}\|_2 + 180\|\mathbf{x} - \mathbf{x}_{[s]}\|_2, \quad \forall t\geq 0.$$

Proof. See the proof in Section 2.5. □

Remark 2.2. *This result appears to be new to the literature, as existing works [6, 4] only established (ℓ_2,ℓ_1) nonuniform instance optimality.*

2.4 Proof of Theorem 2.1

Following the unified framework in [2, Section 2.2.1], the proof architecture is given as follows:

1. **(Decomposition of $\mathbb{S}^{n-1}$)** We let

$$\mathcal{X} = \left\{\mathbf{u}\in\mathbb{S}^{n-1} : \sum_{i\geq 1}\left\|\mathbf{u}_{[(i+1)s]} - \mathbf{u}_{[is]}\right\|_2 \leq \frac{1}{2000}\right\}$$

and accordingly $\mathcal{X}^c := \mathbb{S}^{n-1}\setminus\mathcal{X}$.

2. **(Instance Optimality for $\mathbf{x} \in \mathcal{X}$)**

   RAIC. We establish the signal-dependent RAIC

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u}) \sim \mathrm{RAIC}\left(\Sigma_{2s}^{n,*}; \Sigma_s^{n,*} \cap \mathbb{B}_2^n\left(\mathbf{x}; \frac{1}{20}\right), \mathbf{x}, \frac{\|\mathbf{u}-\mathbf{x}\|_2}{5} + \frac{16}{5}\sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2\right), \quad \forall \mathbf{x} \in \mathcal{X}. \tag{2.6}$$

   By definition, this is equivalent to

$$\|\mathbf{u}-\mathbf{x}-\mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \leq \frac{\|\mathbf{u}-\mathbf{x}\|_2}{5} + \frac{16}{5}\sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2 \quad \forall \mathbf{u} \in \Sigma_s^{n,*} \cap \mathbb{B}_2^n\left(\mathbf{x}; \frac{1}{20}\right), \quad \forall \mathbf{x} \in \mathcal{X}. \tag{2.7}$$

   Under $\|\mathbf{x}-\mathbf{x}_{[s]}\|_2 \leq \sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2 \leq \frac{1}{2000}$ by the construction of $\mathcal{X}$, the conditions in the framework of [2] are satisfied with

$$\mu_1 = \frac{1}{5}, \quad \mu_2 = 0, \quad e(\mathbf{x}) = \frac{16}{5}\sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2$$

   and

$$R_{\mathbf{x}}^{\mathrm{loc}} = \frac{1}{20} > \frac{4e(\mathbf{x}) + 6\|\mathbf{x}-\mathbf{x}_{[s]}\|_2}{1-4\mu_1}.$$

   Initialization. Moreover, we show that $\mathbf{x}_0$ in (2.5) satisfies

$$\|\mathbf{x}_0 - \mathbf{x}\|_2 \leq \frac{1}{20}, \quad \forall \mathbf{x} \in \mathcal{X}. \tag{2.8}$$

   with high probability.

   These components, combined with $\|\mathbf{x}-\mathbf{x}_{[s]}\|_2 \leq \sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2$ and $\|\mathbf{x}_t - \mathbf{x}\|_2 \leq 2$, establish

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq 2\left(\frac{4}{5}\right)^t + 20\sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2, \quad \forall t \geq 0, \ \forall \mathbf{x} \in \mathcal{X}.$$

3. **(Instance Optimality for $\mathbf{x} \in \mathcal{X}^c$)** The main observation is that $\mathbf{x} \notin \mathcal{X}^c$ satisfies $\sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2 > \frac{1}{2000}$, hence we have

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \leq 2 \leq 4000\sum_{i\geq 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2.$$

Overview of the techniques. We focus on the RAIC and the initialization for $\mathbf{x} \in \mathcal{X}$. The main hurdle in the derivation of (2.6) is to bound

$$\sup_{\mathbf{x}\in\mathcal{X}} \sup_{\mathbf{u}\in\Sigma_{4s}^{n,*}} \left| \left\| \frac{\Im(\mathrm{diag}(\overline{\mathrm{sign}(\mathbf{\Phi x})})\mathbf{\Phi u}}{\sqrt{m}} \right\|_2^2 - \|(\mathbf{I} - \mathbf{x}\mathbf{x}^\top)\mathbf{u}\|_2^2 \right|,$$

which is challenging due to the supremum over $\mathbf{x} \in \mathcal{X}$ and the discontinuous $\mathrm{sign}(\mathbf{\Phi x})$-dependence on $\mathbf{x}$. Fortunately, this has been addressed in the orthogonal term analysis in [4]; see Lemma 2.1. In the PBP initialization, by arguments similar to [7, 9], the main error term is (let $\kappa = \sqrt{\pi/2}$)

$$\sup_{\mathbf{x}\in\mathcal{X}} \sup_{\mathbf{w}\in\Sigma_{2s}^{n,*}} \left| \frac{1}{\kappa m}\sum_{i=1}^m \Re(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*\mathbf{u}) - \mathbf{x}^\top\mathbf{u} \right|,$$

which is also bounded by part of the analyses of [4]; see Lemma 2.2.

2.4.1 Proof of the RAIC (2.7)

We start with $(\mathbf{u},\mathbf{x})\in\Sigma_s^{n,*}\times\mathcal{X}$ and leverage the decomposition

$$\begin{aligned}\|\mathbf{u}-\mathbf{x}-\mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2}&=\Big\|\mathbf{u}-\mathbf{x}-\frac{1}{m}\sum_{i=1}^m\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{u}\Big\|_{2s,2}\\&\leq\underbrace{\Big\|\mathbf{u}-\mathbf{x}-\frac{1}{m}\sum_{i=1}^m\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top(\mathbf{u}-\mathbf{x}_{[s]})\Big\|_{2s,2}}_{:=T_1}+\underbrace{\Big\|\frac{1}{m}\sum_{i=1}^m\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{x}_{[s]})\Big\|_{2s,2}}_{:=T_2}.\end{aligned}\tag{2.9}$$

Let $\mathbf{P}_{\mathbf{x}}^\perp=\mathbf{I}_n-\mathbf{x}\mathbf{x}^\top$.

Bounding T_1

Note that it can be further decomposed as follows:

$$\begin{aligned}T_1&\leq\Big\|\mathbf{u}-\mathbf{x}-\frac{1}{m}\sum_{i=1}^m\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top(\mathbf{u}-\mathbf{x}_{[s]})\Big\|_{2s,2}\\&\leq\Big\|\mathbf{P}_{\mathbf{x}}^\perp(\mathbf{u}-\mathbf{x}_{[s]})-\frac{1}{m}\sum_{i=1}^m\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top(\mathbf{u}-\mathbf{x}_{[s]})\Big\|_{2s,2}+\big\|\mathbf{u}-\mathbf{x}-\mathbf{P}_{\mathbf{x}}^\perp(\mathbf{u}-\mathbf{x}_{[s]})\big\|_2:=T_{11}+T_{12}.\end{aligned}$$

For T_{11},

$$\begin{aligned}T_{11}&=\|\mathbf{u}-\mathbf{x}_{[s]}\|_2\cdot\sup_{\mathbf{v}\in\Sigma_{2s}^{n,*}}\Big(\frac{1}{m}\sum_{i=1}^m\mathbf{v}^\top\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top\frac{\mathbf{u}-\mathbf{x}_{[s]}}{\|\mathbf{u}-\mathbf{x}_{[s]}\|_2}-\mathbf{v}^\top\mathbf{P}_{\mathbf{x}}^\perp\frac{\mathbf{u}-\mathbf{x}_{[s]}}{\|\mathbf{u}-\mathbf{x}_{[s]}\|_2}\Big)\\&\leq\|\mathbf{u}-\mathbf{x}_{[s]}\|_2\sup_{\mathbf{u},\mathbf{v}\in\Sigma_{2s}^{n,*}}\Big(\frac{1}{m}\sum_{i=1}^m\langle\mathbf{a}_{i,\mathbf{x}},\mathbf{u}\rangle\langle\mathbf{a}_{i,\mathbf{x}},\mathbf{v}\rangle-\langle\mathbf{P}_{\mathbf{x}}^\perp\mathbf{u},\mathbf{P}_{\mathbf{x}}^\perp\mathbf{v}\rangle\Big):=T_{11}'\cdot\|\mathbf{u}-\mathbf{x}_{[s]}\|_2,\\&\qquad\blacktriangleright\text{ by }\frac{\mathbf{u}-\mathbf{x}_{[s]}}{\|\mathbf{u}-\mathbf{x}_{[s]}\|_2}\in\Sigma_{2s}^{n,*}\ (\text{suppose }\mathbf{u}\neq\mathbf{x}_{[s]})\end{aligned}$$

where

$$\begin{aligned}T_{11}'&=\sup_{\mathbf{u},\mathbf{v}\in\Sigma_{2s}^{n,*}}\Big(\frac{1}{4m}\sum_{i=1}^m\big(\|\mathbf{a}_{i,\mathbf{x}}^\top(\mathbf{u}+\mathbf{v})\|_2^2-\|\mathbf{a}_{i,\mathbf{x}}^\top(\mathbf{u}-\mathbf{v})\|_2^2\big)-\frac{\|\mathbf{P}_{\mathbf{x}}^\perp(\mathbf{u}+\mathbf{v})\|_2^2-\|\mathbf{P}_{\mathbf{x}}^\perp(\mathbf{u}-\mathbf{v})\|_2^2}{4}\Big)\\&\qquad\blacktriangleright\text{ by polar identity}\\&\leq\frac{1}{4}\sup_{\mathbf{w},\mathbf{w}'\in\Sigma_{4s}^n\cap2\mathbb{B}_2^n}\Big(\Big[\frac{1}{m}\sum_{i=1}^m\|\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{w}\|_2^2-\|\mathbf{P}_{\mathbf{x}}^\perp\mathbf{w}\|_2^2\Big]-\Big[\frac{1}{m}\sum_{i=1}^m\|\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{w}'\|_2^2-\|\mathbf{P}_{\mathbf{x}}^\perp\mathbf{w}'\|_2^2\Big]\Big)\\&\qquad\blacktriangleright\text{ by }\mathbf{u}+\mathbf{v},\mathbf{u}-\mathbf{v}\in\Sigma_{4s}^n\cap2\mathbb{B}_2^n\ (\text{when }\mathbf{u},\mathbf{v}\in\Sigma_{2s}^{n,*})\\&\leq2\cdot\frac{1}{4}\sup_{\mathbf{w}\in\Sigma_{4s}^n\cap2\mathbb{B}_2^n}\Big|\frac{1}{m}\sum_{i=1}^m\|\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{w}\|_2^2-\|\mathbf{P}_{\mathbf{x}}^\perp\mathbf{w}\|_2^2\Big|\\&\qquad\blacktriangleright\text{ by }\sup(A-B)\leq\sup|A|+\sup|B|\\&\leq2\sup_{\mathbf{w}\in\Sigma_{4s}^{n,*}}\Big|\frac{1}{m}\sum_{i=1}^m\|\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{w}\|_2^2-\|\mathbf{P}_{\mathbf{x}}^\perp\mathbf{w}\|_2^2\Big|\\&\qquad\blacktriangleright\text{ by homogeneity}\end{aligned}$$

$$= 2 \sup_{\mathbf{w} \in \Sigma_{4s}^{n,*}} \left| \left\| \frac{\Im(\mathrm{diag}(\overline{\mathrm{sign}(\mathbf{\Phi x})})\mathbf{\Phi})\mathbf{w}}{\sqrt{m}} \right\|_2^2 - \|\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{w}\|_2^2 \right|. \tag{2.10}$$

▶ by substituting $\mathbf{a}_{i,\mathbf{x}} = \Im(\overline{\mathrm{sign}(\mathbf{\Phi}_i^* \mathbf{x})}\mathbf{\Phi}_i^*)^\top$

We notice that this is precisely the empirical process analyzed in Appendix A of [4].

Lemma 2.1 (Orthogonal term analysis in [4])**.** *If $m \gtrsim s \log \frac{en}{s}$, then $T'_{11} < \frac{1}{10}$ holds for all $\mathbf{x} \in \mathcal{X}$ with probability at least $1 - C' \exp(-c'm)$.*

Proof. In view of [4, Equation (A.19)], by using the notation therein, (2.10) yields

$$T'_{11} \le 2 \sup_{\mathbf{x} \in \mathcal{X}} \sup_{\mathbf{u} \in \Sigma_{4s}^{n,*}} f_1^{\perp}(\mathbf{x}, \mathbf{u}, 0), \quad \forall \mathbf{x} \in \mathcal{X}.$$

In light of $\omega^2(\mathcal{X}) \lesssim s \log(\frac{en}{s})$ [1] and Sudakov's inequality, [4, Lemma A.7] with the parameter η being a small enough universal constant yields the claim. □

For T_{12}, some algebra finds

$$\begin{aligned}
T_{12} := & \left\|\mathbf{u} - \mathbf{x} - (\mathbf{I}_n - \mathbf{x}\mathbf{x}^\top)(\mathbf{u} - \mathbf{x}_{[s]})\right\|_2 \\
= & \left\|\mathbf{x}\mathbf{x}^\top(\mathbf{u} - \mathbf{x}_{[s]}) - (\mathbf{x} - \mathbf{x}_{[s]})\right\|_2 \\
\le & \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + |\mathbf{x}^\top(\mathbf{u} - \mathbf{x}_{[s]})| \\
\le & \|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + |\mathbf{x}^\top(\mathbf{u} - \mathbf{x})| + |\mathbf{x}^\top(\mathbf{x} - \mathbf{x}_{[s]})| \\
\le & 2\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \frac{\|\mathbf{u} - \mathbf{x}\|_2^2}{2}.
\end{aligned}$$

Combining the pieces together yields

$$\begin{aligned}
T_1 \le & \frac{1}{10}\|\mathbf{u} - \mathbf{x}_{[s]}\|_2 + 2\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \frac{\|\mathbf{u} - \mathbf{x}\|_2^2}{2} \\
\le & \frac{1}{10}\|\mathbf{u} - \mathbf{x}\|_2 + \frac{21}{10}\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 + \frac{\|\mathbf{u} - \mathbf{x}\|_2^2}{2}, \quad \forall (\mathbf{u}, \mathbf{x}) \in \Sigma_s^{n,*} \times \mathcal{X}.
\end{aligned} \tag{2.11}$$

Bounding T_2

We note a useful relation that

$$\mathbf{a}_{i,\mathbf{x}}^\top \mathbf{x} = \Im\big(\overline{\mathrm{sign}(\mathbf{\Phi}_i^* \mathbf{x})}\mathbf{\Phi}_i^*\big)^\top \mathbf{x} = \Im(|\mathbf{\Phi}_i^* \mathbf{x}|) = 0. \tag{2.12}$$

In turn,

$$\begin{aligned}
T_2 = & \left\| \frac{1}{m} \sum_{i=1}^m \mathbf{a}_{i,\mathbf{x}} \mathbf{a}_{i,\mathbf{x}}^\top (\mathbf{x} - \mathbf{x}_{[s]}) \right\|_{2s,2} \\
\le & \sum_{i \ge 1} \left\| \frac{1}{m} \sum_{i=1}^m \mathbf{a}_{i,\mathbf{x}} \mathbf{a}_{i,\mathbf{x}}^\top (\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}) \right\|_{2s,2} \\
\le & \sum_{i \ge 1} \left\| \mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]} \right\|_2 \sup_{\mathbf{u}, \mathbf{v} \in \Sigma_{2s}^{n,*}} \frac{1}{m} \sum_{i=1}^m \mathbf{u}^\top \mathbf{a}_{i,\mathbf{x}} \mathbf{a}_{i,\mathbf{x}}^\top \mathbf{v},
\end{aligned} \tag{2.13}$$

[1] Here, $\omega(\cdot)$ denotes the Gaussian width; see [2, Appendix B].

and it remains to bound $\sup_{\mathbf{x}\in\mathcal{X}}\sup_{\mathbf{u},\mathbf{v}\in\Sigma_{2s}^{n,*}}\frac{1}{m}\sum_{i=1}^m \mathbf{u}^\top\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{v}$. On the event of Lemma 2.1,

$$\begin{aligned} T'_{11} < \frac{1}{10}, \quad \forall \mathbf{x}\in\mathcal{X} &\iff \sup_{\mathbf{x}\in\mathcal{X}}\sup_{\mathbf{u},\mathbf{v}\in\Sigma_{2s}^{n,*}}\left|\frac{1}{m}\sum_{i=1}^m \mathbf{u}^\top\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{v} - \langle \mathbf{P}_{\mathbf{x}}^\perp\mathbf{u}, \mathbf{P}_{\mathbf{x}}^\perp\mathbf{v}\rangle\right| \le \frac{1}{10} \\ &\implies \sup_{\mathbf{x}\in\mathcal{X}}\sup_{\mathbf{u},\mathbf{v}\in\Sigma_{2s}^{n,*}}\frac{1}{m}\sum_{i=1}^m \mathbf{u}^\top\mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{v} \le \frac{11}{10}. \end{aligned}$$

Substituting this into (2.13) establishes

$$T_2 \le \frac{11}{10}\sum_{i\ge 1}\left\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\right\|_2, \quad \forall \mathbf{x}\in\mathcal{X}. \tag{2.14}$$

Putting Pieces Together

By substituting (2.11) and (2.14) into (2.9) and using $\|\mathbf{x}-\mathbf{x}_{[s]}\|_2 \le \sum_{i\ge 1}\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\|_2$, we obtain for all $(\mathbf{u},\mathbf{x})\in\Sigma_s^{n,*}\times\mathcal{X}$ the bound

$$\|\mathbf{u}-\mathbf{x}-\mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \le \frac{\|\mathbf{u}-\mathbf{x}\|_2}{10} + \frac{\|\mathbf{u}-\mathbf{x}\|_2^2}{2} + \frac{16}{5}\sum_{i\ge 1}\left\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\right\|_2.$$

Therefore, by restricting $\mathbf{u}$ to $\Sigma_s^{n,*}\cap\mathbb{B}_2^n(\mathbf{x};\frac{1}{20})$ for each $\mathbf{x}$,

$$\|\mathbf{u}-\mathbf{x}-\mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} \le \frac{\|\mathbf{u}-\mathbf{x}\|_2}{5} + \frac{16}{5}\sum_{i\ge 1}\left\|\mathbf{x}_{[(i+1)s]} - \mathbf{x}_{[is]}\right\|_2,$$

as desired.

2.4.2 Proof of the Initialization Bound (2.8)

In view of

$$\mathbb{E}\left[\Re\left(\frac{1}{m}\sum_{i=1}^m z_i\mathbf{\Phi}_i\right)\right] = \Re\Big(\mathbb{E}\big[\operatorname{sign}(\mathbf{\Phi}_i^*\mathbf{x})\mathbf{\Phi}_i\big]\Big) = \mathbb{E}\big[\operatorname{sign}(\mathbf{\Phi}_i^*\mathbf{x})\mathbf{x}^\top\mathbf{\Phi}_i\mathbf{x}\big] = \mathbb{E}\big[|\mathbf{\Phi}_i^*\mathbf{x}|\mathbf{x}\big] = \sqrt{\frac{\pi}{2}}\mathbf{x} := \kappa\mathbf{x},$$

we have

$$\begin{aligned} \|\mathbf{x}_0 - \mathbf{x}\|_2 &= \left\|\frac{H_s(\Re(\frac{1}{m}\sum_{i=1}^m z_i\mathbf{\Phi}_i))}{\|H_s(\Re(\frac{1}{m}\sum_{i=1}^m z_i\mathbf{\Phi}_i))\|_2} - \frac{\kappa\mathbf{x}}{\|\kappa\mathbf{x}\|_2}\right\|_2 \\ &\le \frac{2}{\kappa}\left\|H_s\left(\Re\left(\frac{1}{m}\sum_{i=1}^m z_i\mathbf{\Phi}_i\right)\right) - \kappa\mathbf{x}\right\|_2 \\ &\qquad \blacktriangleright \text{ by, e.g., [4, Equation (2.12)]} \\ &\le \frac{2}{\kappa}\left\|H_s\left(\Re\left(\frac{1}{m}\sum_{i=1}^m z_i\mathbf{\Phi}_i\right)\right) - \kappa\mathbf{x}_{[s]}\right\|_2 + 2\|\mathbf{x}-\mathbf{x}_{[s]}\|_2 \\ &\qquad \blacktriangleright \text{ by triangle inequality} \\ &\le \frac{4}{\kappa}\left\|\frac{1}{m}\sum_{i=1}^m \Re(z_i\mathbf{\Phi}_i) - \kappa\mathbf{x}_{[s]}\right\|_{2s,2} + 2\|\mathbf{x}-\mathbf{x}_{[s]}\|_2 \end{aligned}$$

▶ by [2, Lemma A.1]

$$\leq 4\left\|\frac{1}{\kappa m}\sum_{i=1}^{m}\Re(z_i\mathbf{\Phi}_i)-\mathbf{x}\right\|_{2s,2}+6\|\mathbf{x}-\mathbf{x}_{[s]}\|_2. \tag{2.15}$$

▶ by triangle inequality

The first term can be controlled by the parallel term analysis in Appendix A of [4].

Lemma 2.2 (Parallel term analysis in [4])**.** *If $m \gtrsim s\log(\frac{en}{s})$, then*

$$\sup_{\mathbf{x}\in\mathcal{X}}\left\|\frac{1}{\kappa m}\sum_{i=1}^{m}\Re(z_i\mathbf{\Phi}_i)-\mathbf{x}\right\|_{2s,2}\leq\frac{1}{160}$$

holds with probability at least $1-\exp(-c'm)$.

Proof. By using the notation in Equation (A.15) of [4],

$$\sup_{\mathbf{x}\in\mathcal{X}}\left\|\frac{1}{\kappa m}\sum_{i=1}^{m}\Re(z_i\mathbf{\Phi}_i)-\mathbf{x}\right\|_{2s,2}=\sup_{\mathbf{x}\in\mathcal{X}}\sup_{\mathbf{u}\in\Sigma_{2s}^{n,*}}\frac{1}{\kappa m}\sum_{i=1}^{m}\Re(\overline{\mathrm{sign}(\mathbf{\Phi}_i^*\mathbf{x})}\mathbf{\Phi}_i^*\mathbf{u})-\mathbf{x}^\top\mathbf{u}\leq\sup_{\mathbf{x}\in\mathcal{X}}\sup_{\mathbf{u}\in\Sigma_{2s}^{n,*}}|f_1^{\|}(\mathbf{x},\mathbf{u})|.$$

Using $\omega^2(\mathcal{X})\leq s\log\frac{en}{s}$ and Sudakov's inequality, [4, Lemma A.6] with the parameter η being a small enough universal constant yields the claim. □

In light of (2.15), for any $\mathbf{x}\in\mathcal{X}$,

$$\|\mathbf{x}_0-\mathbf{x}\|_2\leq 4\sup_{\mathbf{x}\in\mathcal{X}'}\left\|\frac{1}{\kappa m}\sum_{i=1}^{m}\Re(z_i\mathbf{\Phi}_i)-\mathbf{x}\right\|_{2s,2}+6\|\mathbf{x}-\mathbf{x}_{[s]}\|_2<\frac{4}{160}+\frac{6}{2000}\leq\frac{1}{20},$$

as desired.

2.5 Proof of Theorem 2.2

We follow the unified framework in [2, Section 2.2.2] to prove it:

1. **(Decomposition of $\mathbb{S}^{n-1}$)** We let $\mathcal{X}=\big\{\mathbf{u}\in\mathbb{S}^{n-1}:\|\mathbf{u}-\mathbf{u}_{[s]}\|_2\leq\frac{1}{90}\big\}$.

2. **(Instance Optimality if $\mathbf{x}\in\mathcal{X}$)** If $\mathbf{x}\in\mathcal{X}$, we prove for the specific $\mathbf{x}$ that the RAIC

$$\mathbf{h}_{\mathbf{x}}(\mathbf{u})\sim\mathrm{RAIC}\bigg(\Sigma_s^n;\Sigma_s^{n,*}\cap\mathbb{B}_2^n\Big(\mathbf{x};\frac{1}{3}\Big),\mathbf{x},\frac{\|\mathbf{u}-\mathbf{x}\|_2}{5}\bigg) \tag{2.16}$$

holds with the promised probability. By definition, this is equivalent to

$$\|\mathbf{u}-\mathbf{x}-\mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2}\leq\frac{\|\mathbf{u}-\mathbf{x}\|_2}{5},\quad\forall\mathbf{u}\in\Sigma_s^{n,*}\cap\mathbb{B}_2^n\Big(\mathbf{x},\frac{1}{3}\Big). \tag{2.17}$$

This RAIC renders the conditions in the framework with $\mu_1=\frac{1}{5}$, $\mu_2=e(\mathbf{x})=0$, $R_{\mathbf{x}}^{\mathrm{loc}}=\frac{1}{3}$, and note that

$$R_{\mathbf{x}}^{\mathrm{loc}}=\frac{1}{3}>30\|\mathbf{x}-\mathbf{x}_{[s]}\|_2>\frac{6\|\mathbf{x}-\mathbf{x}_{[s]}\|_2}{1-4\mu_1}$$

Moreover, we show that

$$\mathbf{x}_0\in\Sigma_s^{n,*}\cap\mathbb{B}_2^n(\mathbf{x};\frac{1}{3}) \tag{2.18}$$

holds with the promised probability. Therefore, the framework yields

$$\|\mathbf{x}_t-\mathbf{x}\|_2\leq 2\Big(\frac{4}{5}\Big)^t+30\|\mathbf{x}-\mathbf{x}_{[s]}\|_2.$$

3. **(Instance Optimality if $\mathbf{x} \in \mathcal{X}^c$)** If $\mathbf{x} \in \mathcal{X}^c$, then $\|\mathbf{x} - \mathbf{x}_{[s]}\|_2 > \frac{1}{20}$, and hence we always have

$$\|\mathbf{x}_t - \mathbf{x}\|_2 \le 2 \le 180\|\mathbf{x} - \mathbf{x}_{[s]}\|_2.$$

Overview of the techniques. We focus on the RAIC in (2.16) that is sharper than the previous (2.6). The main observation is that the $\mathrm{sign}(\mathbf{\Phi x})$-dependence is not problematic if we only consider a fixed $\mathbf{x}$. For instance, by conditioning on $\mathrm{sign}(\mathbf{\Phi}_i^* \mathbf{x})$ and the rotational invariance, $\mathbf{a}_{i,\mathbf{x}}$ defined in (2.3) behaves as a standard Gaussian vector over $(\mathbf{I}_n - \mathbf{x}\mathbf{x}^\top)\mathbb{R}^n$. In turn, with some linear algebra, the derivation of (2.16) essentially reduces to a standard proof of the RIP of a Gaussian matrix, which we accomplish through [2, Lemma C.13] (concentration of product process).

2.5.1 Proof of the RAIC (2.17)

For the fixed $\mathbf{x}$ and any $\mathbf{u} \in \Sigma_s^{n,*}$,

$$\begin{aligned}
\|\mathbf{u} - \mathbf{x} - \mathbf{h}_{\mathbf{x}}(\mathbf{u})\|_{2s,2} &= \left\|\mathbf{u} - \mathbf{x} - \frac{1}{m}\sum_{i=1}^m \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \mathbf{u}\right\|_{2s,2} \\
&= \left\|\langle \mathbf{u} - \mathbf{x}, \mathbf{x}\rangle \mathbf{x} + \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u} - \frac{1}{m}\sum_{i=1}^m \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\right\|_{2s,2} \\
&\qquad \blacktriangleright \text{ by } \mathbf{u} - \mathbf{x} = \langle \mathbf{u} - \mathbf{x}, \mathbf{x}\rangle \mathbf{x} + \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u} \text{ and Equation (2.12)} \\
&\le \frac{\|\mathbf{u} - \mathbf{x}\|_2^2}{2} + \left\|\frac{1}{m}\sum_{i=1}^m \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u} - \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\right\|_{2s,2}. \qquad (2.19) \\
&\qquad \blacktriangleright \text{ by triangle inequality and } \|\langle \mathbf{u} - \mathbf{x}, \mathbf{x}\rangle \mathbf{x}\|_{2s,2} = \frac{\|\mathbf{u} - \mathbf{x}\|_2^2}{2}
\end{aligned}$$

Also, the second term can be bounded by

$$\begin{aligned}
&\left\|\frac{1}{m}\sum_{i=1}^m \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u} - \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\right\|_{2s,2} = \sup_{\mathbf{v} \in \Sigma_{2s}^{n,*}} \frac{1}{m}\sum_{i=1}^m \mathbf{v}^\top \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u} - \mathbf{v}^\top \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u} \\
&= \|\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\|_2 \cdot \sup_{\mathbf{v} \in \Sigma_{2s}^{n,*}} \frac{1}{m}\sum_{i=1}^m (\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{v})^\top \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \left(\frac{\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}}{\|\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\|_2}\right) - \left\langle \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{v}, \frac{\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}}{\|\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\|_2}\right\rangle \\
&\qquad \blacktriangleright \text{ by Equation (2.12)} \\
&= \|\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\|_2 \cdot \sup_{\mathbf{v} \in \Sigma_{2s}^{n,*}} \left\{\frac{1}{m}\sum_{i=1}^m (\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{v})^\top \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \left(\frac{\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}}{\|\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\|_2}\right) - \mathbb{E}\left[(\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{v})^\top \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \left(\frac{\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}}{\|\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\|_2}\right)\right]\right\} \\
&\qquad \blacktriangleright \text{ by rotation invariance, } \mathbf{a}_{i,\mathbf{x}} \text{ essentially behaves as } N(0, \mathbf{I}_{n-1}) \text{ over } P_{\mathbf{x}}^{\perp}\mathbb{R}^n \\
&\le \|\mathbf{u} - \mathbf{x}\|_2 \cdot \underbrace{\sup_{\mathbf{v} \in \mathbf{P}_{\mathbf{x}}^{\perp}\Sigma_{2s}^{n,*}} \sup_{\mathbf{w} \in (\Sigma_s^n - \mathbb{R}\mathbf{x}) \cap \mathbb{S}^{n-1}} \left\{\frac{1}{m}\sum_{i=1}^m \mathbf{v}^\top \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \mathbf{w} - \mathbb{E}\big[\mathbf{v}^\top \mathbf{a}_{i,\mathbf{x}}\mathbf{a}_{i,\mathbf{x}}^\top \mathbf{w}\big]\right\}}_{:=T}. \qquad (2.20) \\
&\qquad \blacktriangleright \text{ by } \|\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\|_2 \le \|\mathbf{u} - \mathbf{x}\|_2,\ \mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{v} \in \mathbf{P}_{\mathbf{x}}^{\perp}\Sigma_{2s}^{n,*},\ \frac{\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}}{\|\mathbf{P}_{\mathbf{x}}^{\perp}\mathbf{u}\|_2} = \frac{\mathbf{u} - (\mathbf{x}^\top \mathbf{u})\mathbf{x}}{\|\mathbf{u} - (\mathbf{x}^\top \mathbf{u})\mathbf{x}\|_2} \in (\Sigma_s^n - \mathbb{R}\mathbf{x}) \cap \mathbb{S}^{n-1}
\end{aligned}$$

To bound T, we notice that for a fixed $\mathbf{x}$,

$$\|\mathbf{a}_{i,\mathbf{x}}^\top\|_{\psi_2} = \sup_{\mathbf{w} \in \mathbb{S}^{n-1}} \left\|\Im(\overline{\mathrm{sign}(\mathbf{\Phi}_i^* \mathbf{x})\mathbf{\Phi}_i^*})^\top \mathbf{w}\right\|_{\psi_2} \le \sup_{\mathbf{w} \in \mathbb{S}^{n-1}} \left\||\mathbf{\Phi}_i^* \mathbf{w}|\right\|_{\psi_2}$$

$$\le \sup_{\mathbf{w}\in\mathbb{S}^{n-1}} \left\|\Re(\mathbf{\Phi}_i^*\mathbf{w})\right\|_{\psi_2} + \sup_{\mathbf{w}\in\mathbb{S}^{n-1}} \left\|\Im(\mathbf{\Phi}_i^*\mathbf{w})\right\|_{\psi_2} \lesssim 1,$$

and hence for any $\mathbf{u},\mathbf{v}\in\mathbb{S}^{n-1}$,

$$\|\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{u}-\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{v}\|_{\psi_2} \lesssim \|\mathbf{u}-\mathbf{v}\|_2\,, \quad \|\mathbf{a}_{i,\mathbf{x}}^\top\mathbf{u}\|_{\psi_2} \lesssim 1.$$

Combining with $\omega(P_{\mathbf{x}}^{\perp}\Sigma_{2s}^{n,*}) \le \omega(\Sigma_{2s}^{n,*}) \lesssim \sqrt{s\log(en/s)}$ and $\omega((\Sigma_s^n-\mathbb{R}\mathbf{x})\cap\mathbb{S}^{n-1}) \lesssim \sqrt{s\log\frac{en}{s}}$,[2] for any $u\ge 0$, [2, Lemma C.13] yields

$$\mathbb{P}\left(T \lesssim \frac{s\log(en/s)+u}{m} + \sqrt{\frac{s\log(en/s)+u}{m}}\right) \ge 1-\exp(-cu).$$

We set $u=c_0m$ with small enough c_0 establishes $T\le\frac{1}{5}$ with the promised probability. Combining with (2.20) and (2.19) and using $\|\mathbf{u}-\mathbf{x}\|_2^2 \le \frac{1}{3}\|\mathbf{u}-\mathbf{x}\|_2$ establish (2.17). The proof is complete.

2.5.2 Proof of the Initialization Bound (2.18)

Note that (2.15) remains valid. For the fixed $\mathbf{x}$, we can use the [4, Lemma A.3] to ensure that

$$\left\|\frac{1}{\kappa m}\sum_{i=1}^m \Re(z_i\mathbf{\Phi}_i)-\mathbf{x}\right\|_{2s,2}$$

is small enough. Further using the definition of $\mathcal{X}$ yields (2.18).

2.6 A Gaussian Width Estimate

For any $\mathbf{u}\in\mathbb{R}^n$, we prove

$$\omega\big((\Sigma_s^n-\mathbb{R}\mathbf{u})\cap\mathbb{S}^{n-1}\big) \lesssim \sqrt{s\log\frac{en}{s}}$$

by estimating the covering number. By writing

$$\Sigma_s^n = \bigcup_{1\le j\le\binom{n}{s}} V_j, \quad \text{where } V_j \text{ are } s\text{-dimensional linear subspace in } \mathbb{R}^n,$$

we arrive at

$$(\Sigma_s^n-\mathbb{R}\mathbf{u})\cap\mathbb{S}^{n-1} = \bigcup_{1\le j\le\binom{n}{s}} (V_j-\mathbb{R}\mathbf{u})\cap\mathbb{S}^{n-1},$$

and notice that $\tilde{V}_j = V_j-\mathbb{R}\mathbf{u}$ are $(s+1)$-dimensional linear subspace. For any $r\in(0,2)$, a standard estimate gives $\mathscr{N}(\tilde{V}_j\cap\mathbb{S}^{n-1},r) \le (\frac{4}{r})^{s+1}$, and therefore

$$\mathscr{N}\big((\Sigma_s^n-\mathbb{R}\mathbf{u})\cap\mathbb{S}^{n-1},r\big) \le \binom{n}{s}\left(\frac{4}{r}\right)^{s+1} \le \left(\frac{en}{s}\right)^s\left(\frac{4}{r}\right)^{s+1} \le \left(\frac{12n}{sr}\right)^{s+1}.$$

By Dudley's inequality (see, e.g., [8, Theorem 8.1.10]),

$$\omega\big((\Sigma_s^n-\mathbb{R}\mathbf{u})\cap\mathbb{S}^{n-1}\big) \lesssim \int_0^2 \sqrt{\log\mathscr{N}\big((\Sigma_s^n-\mathbb{R}\mathbf{u})\cap\mathbb{S}^{n-1},r\big)}\,dr \lesssim \sqrt{s\log\frac{en}{s}}\,.$$

[2] We establish this estimate separately in Section 2.6.

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